Oriental University

RansomRadar

Behaviour-based ransomware early warning

Detect the behaviour.

Not just the fingerprint.

Same threat. A different question.

vs

Signature-based detection

“Have we seen this file before?”

Hashes · digital fingerprints
Known signatures · blocklists

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“What is it doing to the files?”

Writes · renames · file-type changes
Unusual content · canary activity

Ransomware leaves a trail.



Write many files

Change content

Rename extensions

File activity → early warning

Watch	Measure	Detect	Explain
File events	Behaviour + content	Rules + normal baseline	Alert + evidence

Observe the pattern—not the malware's name.

Learn normal. Notice the unusual.

01 Everyday file activity

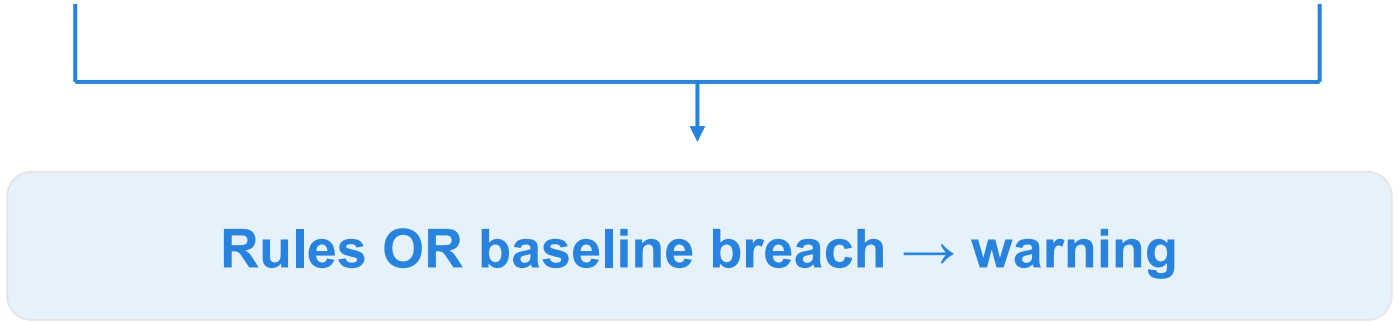
Edits · saves · legitimate copies

Learn what normal work looks like.

02 Suspicious deviation

Writes · renames · timing changes

Look for departures from that baseline.



Entropy is a clue. Not a verdict.

+

Entropy: byte randomness

0–8 bits per byte

Encryption can push it high.
So can legitimate compression.

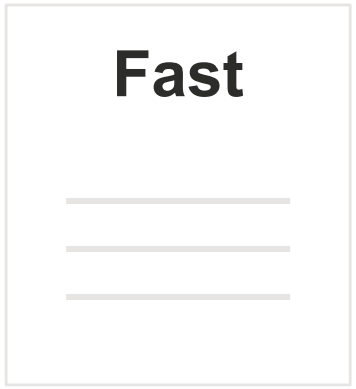
Header: file-format evidence

Does the format still match?

A .docx should retain its ZIP header.
We check a 4 KB content sample.

We built the attacker, too.

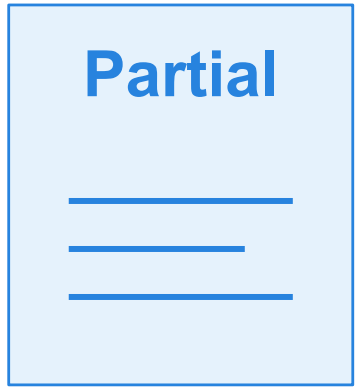
Same sandbox. Three ways to challenge the detector.



Rewrite in bursts



Slow the writes



Corrupt the header

We test the failure mode—not just the happy path.

Entropy missed it. The full detector caught it.

Header-only rewrite · 2 synthetic attack sessions

Only the header changes. File activity still leaves evidence.



0 / 2

Entropy-only

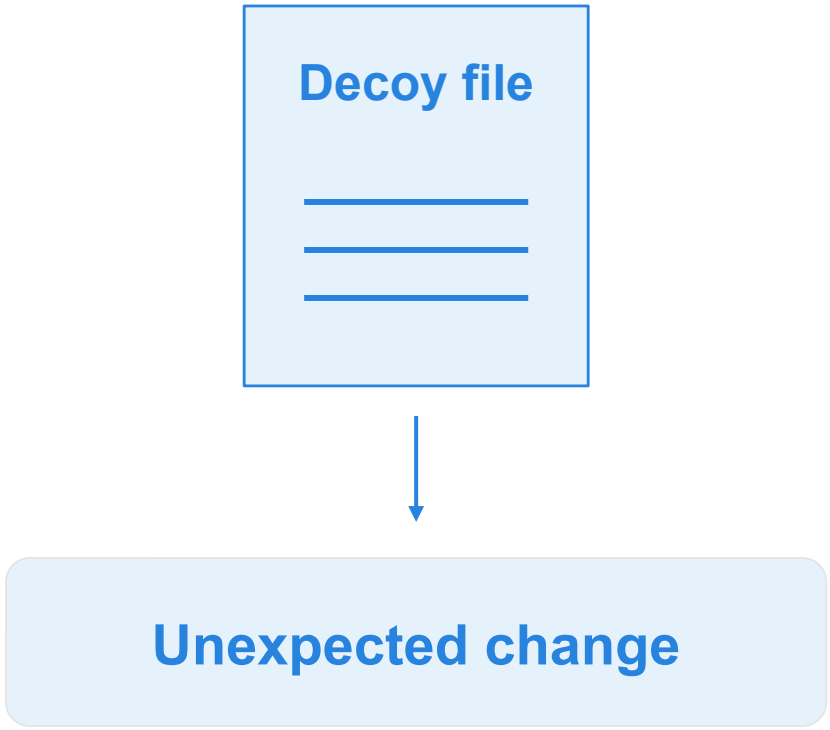
Attack sessions detected

2 / 2

Full learned detector

Attack sessions detected

A decoy file becomes a tripwire.



PLACE

Create a decoy inside the monitored folder.

WATCH

Notice unexpected modification or renaming.

WARN

Raise a high-priority alert with the file evidence.

A score needs context—and a reason.

+

Is this unusual?

ECOD + Isolation Forest

Compare with benign history.
Add a conformal p-value.

Which signals drove it?

Exact top-6 Shapley

Attribute the ensemble score.
Show the strongest feature drivers.

37 features: rates · ratios · timing · trends · content

Not just “suspicious”. Here is why.

WARNING

↓

File evidence

Reason for the alert

CANARY

A protected decoy was touched.

ACTIVITY

504 events/min exceeded the learned allowance.

CONTENT

High entropy + 33 file-type mismatches.

What we built it with

Python	watchdog	scikit-learn	SQLite	Streamlit
Core logic	File events	Anomaly models	Evidence store	Dashboard

Monitoring → analysis → evidence → interface

Judge the warning by what it costs.

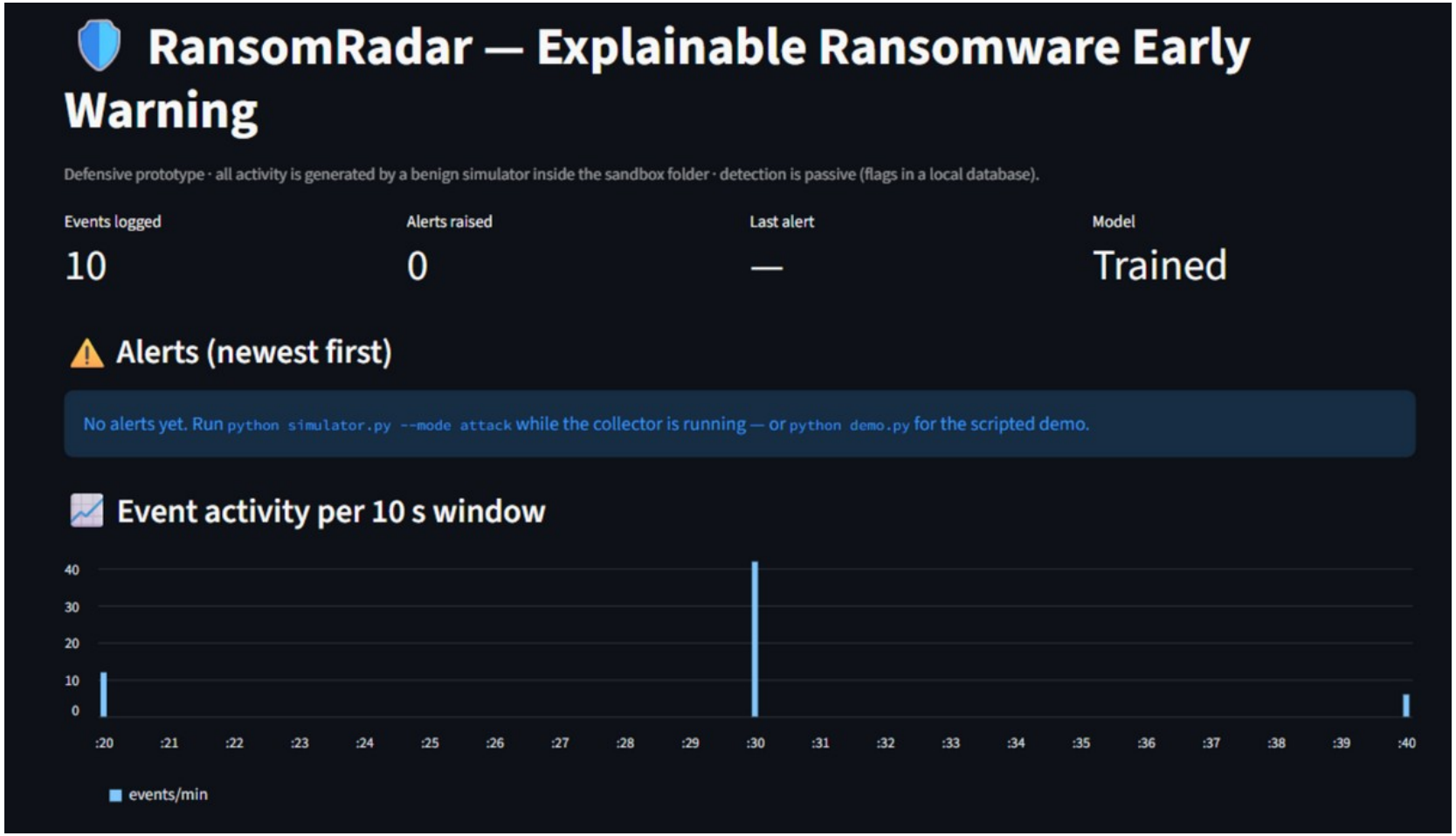
01 **Damage: how many files changed before the warning?**

02 Delay: how quickly did the first warning appear?

03 False alarms: how often does normal work trigger it?

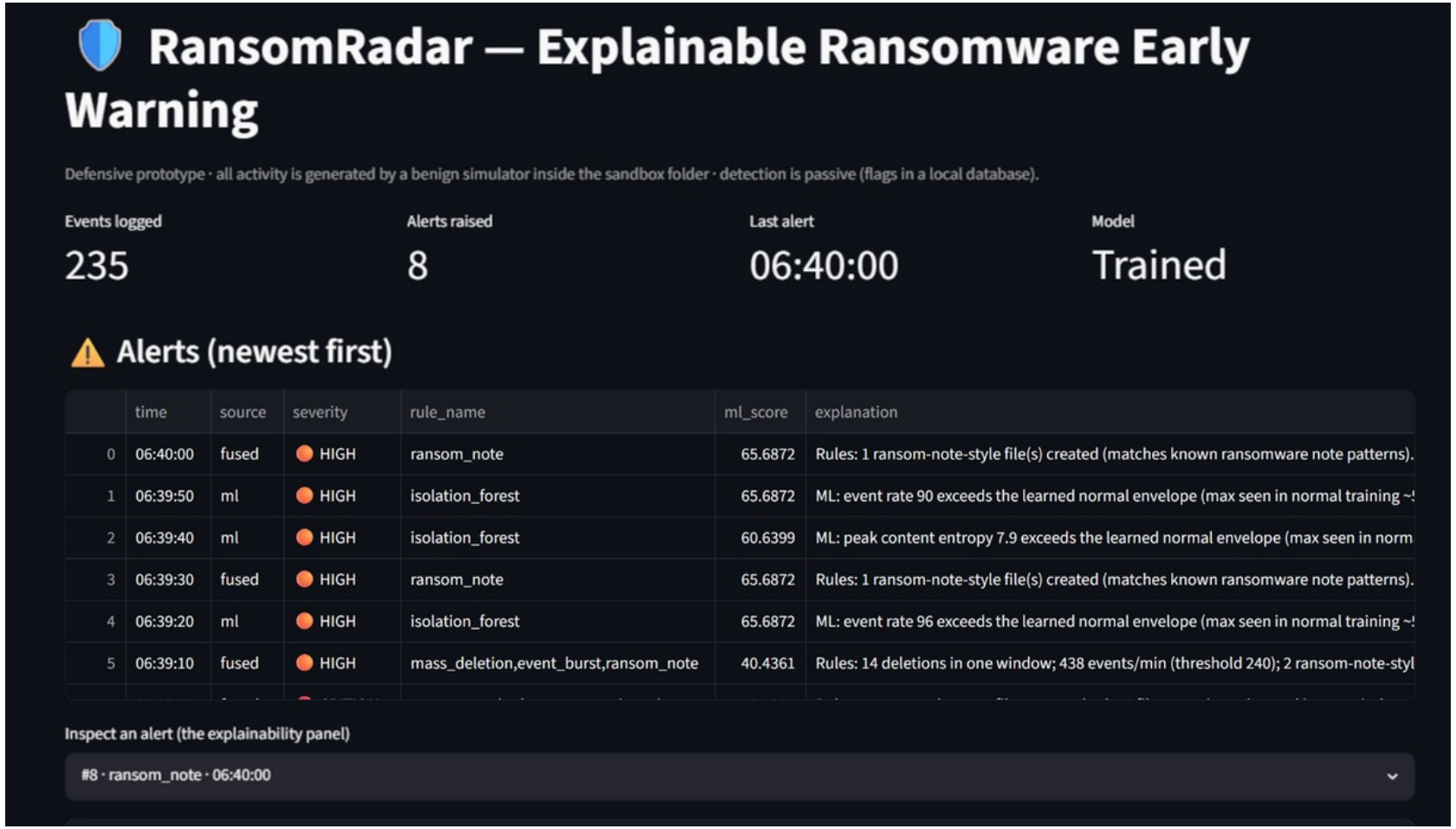
Normal work. No warning.

10 events · 0 alerts · model trained



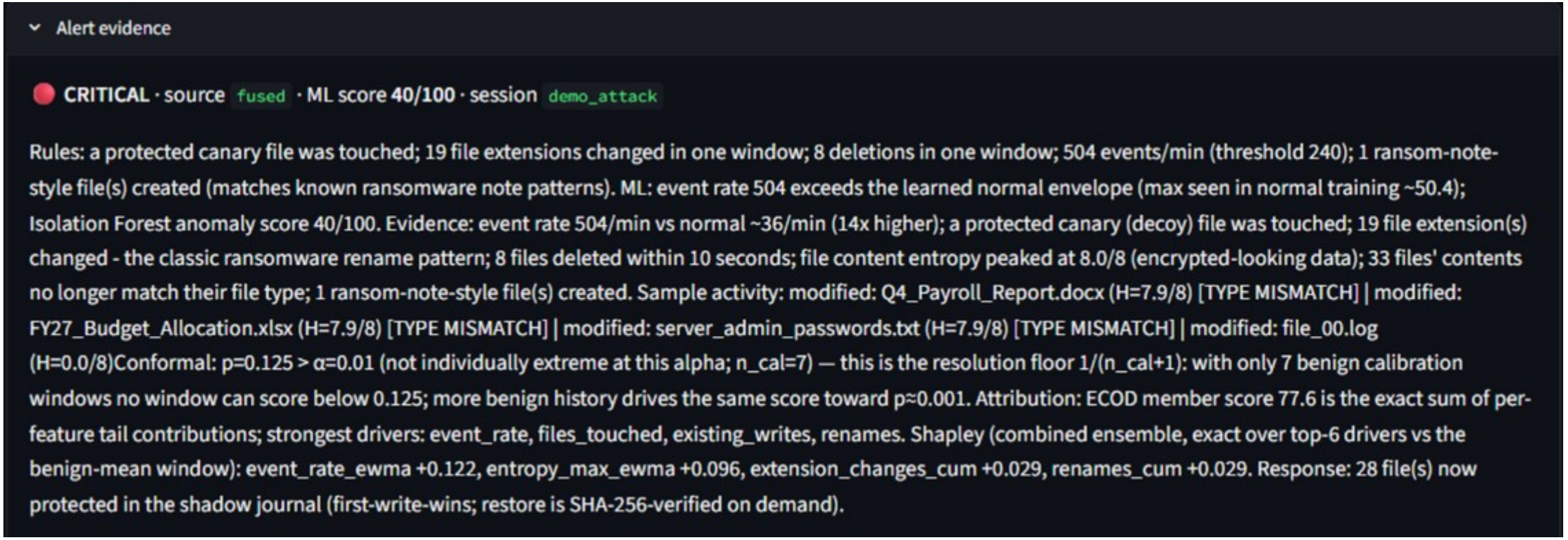
The attack becomes visible.

235 events · 8 alerts · reasons retained



The alert explains itself.

Rules + baseline · calibration context · feature attributions



Recovery, checked byte for byte.



30 / 30 originals restored · SHA-256 verified

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Thank you

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Pratik Goyal · Yash Chaudhary

Guided by Suraj Batham